

Airframe Parameter Comparison

All FC parameters in unified-float format (radians, volts, fractions). **Bold** values are dramatic outliers relative to the airframe set median.

Param	150mm Brushed	DevEBox	Ecks 220mm	Ken's 450	Ken's Alpha	Ken's LadyBug	Phoenix FW
RollRateKp	0.16	0.095	0.1	0.16	0.095	0.16	0.04
AltPosKi	0.0046	0.0046	0.0046	0.00092	0.00184	0.0138	0.00552
RollAngleQKp	7	5.75	7	7	5.75	7	2
RollAngleQIntLimit	0.005236	0.002618	0.002618	0.00733	0.002618	0.005236	0.002618
PitchRateKp	0.16	0.095	0.1	0.16	0.095	0.16	0.04
AltPosKp	0.5124	0.5124	0.5124	0.1464	0.4575	0.915	0.2928
PitchAngleQKp	7	5.75	7	7	5.75	7	2
PitchAngleQIntLimit	0.005236	0.002618	0.002618	0.00733	0.002618	0.005236	0.002618
YawRateKp	0.2	0.125	0.1	0.2	0.125	0.2	0.05
RollRateKd	0.006	0.004	0.0045	0.006	0.004	0.006	0
BbLogType	0	0	0	0	0	0	0
Config1Bits	-	-	-	-	-	-	-
RxThrottleCh	0	0	0	0	0	0	0
LowVoltThres	3.4	9.6	9.6	13.6	13.8	3	10.2
RollCamKp	1	1	1	1	1	1	0
EstCruiseThr	0.59	0.3	0.54	0.31	0.3	0.54	0.2
StickHysteresis	0.02	0.03	0.02	0.04	0.03	0.02	0.02
FwClimbThrottle	0	0	0	0	0	0	0.7
PercentIdleThr	0.1	0.05	0.05	0.1	0.05	0.1	0.05
RollAngleQKi	0.1	0.15	0.25	0.2	0.15	0.1	0.1
PitchAngleQKi	0.1	0.15	0.25	0.1	0.15	0.1	0.1
PitchCamKp	1	1	1	1	1	1	0
ServoLpfHz	25	25	25	10	25	25	25
PitchRateKd	0.006	0.004	0.0045	0.006	0.004	0.006	0
NavVelKp	0.6	0.6	0.3	0.6	0.6	0.6	0.6
Horizon	0.3	0.4	0.3	0.4	0.4	0.3	0.01
MadgwickKpMag	0.5	0.4	0.5	0.5	0.4	0.5	0.5
NavRthAlt	10	10	10	10	10	10	30
NavMagVar	0.204204	0.204204	0.205949	0.226893	0.204204	0.226893	0.204204
RcChannels	7	7	7	7	7	7	7
RxRollCh	1	1	1	1	1	1	1
MadgwickKpAcc	0.2	0.18	0.2	0.2	0.18	0.2	0.2
RollCamTrim	0	0	0	0.01	0	0	0
NavPosIntLimit	8	5	3	5	5	8	12
RxPitchCh	2	2	2	2	2	2	2
RxYawCh	3	3	3	3	3	3	3
MaxDescentRateDmpS	12	30	10	18	30	12	25
DescentDelayS	15	10	15	10	10	15	15
GyroLpfSel	2	2	2	2	2	2	2
NavCrossTrackKp	0.04	0.04	0.04	0.04	0.04	0.04	0.04
RxGearCh	4	4	4	5	5	5	4
RxAux1Ch	5	5	5	8	11	8	5
ServoSense	9	0	0	0	0	9	9
AccConfSd	0.06	0.04	0.06	0.06	0.04	0.06	0.06
BatteryCapacity	22	22	22	22	22	22	22
RxAux2Ch	6	6	6	6	6	6	6

Param	150mm Brushed	DevEBox	Ecks 220mm	Ken's 450	Ken's Alpha	Ken's LadyBug	Phoenix FW
RxAux3Ch	7	10	10	7	7	7	10
NavPosKp	0.33	0.2475	0.2475	0.2475	0.2475	0.33	0.33
AltLpf	10	10	10	20	10	20	10
Balance	0.5	0.5	0.5	0.5	0.5	0.5	0.5
RxAux4Ch	8	8	8	11	8	10	8
NavPosKi	0.008	0.012	0.012	0.012	0.012	0.008	0.008
TiltThrottleFf	0	0.1	0	0.25	0.1	0	0
MaxYawRate	7.67945	4.18879	2.0944	7.67945	5.23599	7.67945	2.0944
FwRollPitchFf	0.25	0	0	0	0	0.25	0
FwPitchThrottleFf	0.1	0	0	0	0	0.1	0
FwMaxClimbAngle	0.261799	1.0472	1.0472	1.0472	1.0472	0.261799	0.523599
NavMaxAngle	0.261799	0.349066	0.436332	0.349066	0.349066	0.261799	0.349066
FwSpoilerDecayPercentPs	0.015	0.01	0.01	0	0	0.015	0.02
FwAileronDifferential	0.3	0	0	0	0	0.3	0
KfAccUBiasVar	3e-06	1e-06	3e-06	5e-06	0	0	2e-06
Config2Bits	-	-	-	-	-	-	-
MaxPitchAngle	1.0472	1.0472	0.523599	1.0472	1.0472	1.0472	1.0472
MaxRollAngle	1.0472	1.0472	0.523599	1.0472	1.0472	1.0472	1.0472
YawLpfHz	30	50	50	50	50	30	50
NavHeadingTurnout	0.698132	0.349066	0.349066	0.349066	0.349066	0.698132	0.349066
AltHoldThrCompDecayPercentPs	0.001	0.02	0.025	0	0.001	0	0.001
FwBoardPitchAngle	0.087266	0	0	0	0	0.087266	0
MaxRollRate	10.472	10.472	10.472	10.472	10.472	10.472	10.472
MaxPitchRate	10.472	10.472	10.472	10.472	10.472	10.472	10.472
CurrentScale	100	100	100	100	100	100	100
VoltScale	100	100	100	100	100	100	100
FwAileronRudderMix	1	0	0	0	0	1	0
FwAltSpoilerFf	0.5	0	0	0	0	0.5	0
MaxCompassYawRate	1.5708	1.0472	1.0472	1.0472	1.0472	1.5708	1.0472
AccLpfSel	4	4	4	4	4	4	4
YawRateKd	0.000125	0.001125	0.001125	0.001125	0.001125	0.000125	0.001125
ThrottleGainRate	0	0	0	0	0	0	0
RxAux5Ch	9	9	9	9	9	9	9
RxAux6Ch	10	7	7	10	10	11	7
RxAux7Ch	11	11	11	4	4	4	11
YawAngleQKp	2.25	3	3	3	3	2.25	3
YawAngleQKi	0.25	0.25	0.25	0.25	0.25	0.25	0.25
YawAngleQIntLimit	0.008727	0.008727	0.008727	0.008727	0.008727	0.008727	0.008727
AltPosIntLimit	0.5	0.5	0.5	0.5	0.5	0.5	0.75
MotorStopSel	0	0	0	0	0	0	0
AltThrottleCompLimit	0.02	0.2	0.2	0.12	0.05	0.02	0.1
VrsRoc	50	30	50	30	30	50	30
AhRocWindowMps	5	15	20	30	5	5	5
NavProxAltM	5	3	4	3	3	5	3
NavProxRadiusM	10	1	3	2	1	10	3
KfBaroVar	50	20	100	50	50	50	50
KfAccUVar	0.2	0.2	0.2	0.2	0.2	0.2	0.2
FwStickScale	4	4	4	2	4	4	4
FwRollControlPitchLimit	0.785398	0.785398	0.785398	0.785398	0.785398	0.785398	0.785398
AhThrottleMovingTrigger	0.3	0.2	0.2	0.11	0.2	0.3	0.2
NavFenceRadiusM	100	100	100	200	100	100	100

Notes

- Outlier detection: values > 3 MAD from the median are bolded.
- Unused/reserved parameters omitted.